Auditing Verification Decisions in Agent Self-Improvement

Anonymous ACL submission

Abstract

Self-improving agent loops change an agent's harness (its prompts, tools and code) and keep a change only when verification evidence says it helps. We audit these decisions one candidate at a time: inside RRSI, a self-improvement loop that edits harness code, run with DeepSeek models on τ^2-bench and AppWorld, we deploy every measured candidate, accepted or rejected, on held-out tasks and account for what its evidence cost. Most candidate-level effects remained unresolved: unadjusted 95% intervals excluded zero positively for 8 of the 183 measured candidates and negatively for 12. Evidence that never ran the changed agent on solved tasks would have accepted changes that crashed the harness or blocked legitimate actions, and no rule's verification was shown to pay for itself against keeping the incumbent. A cheaper rule that stops evaluating a candidate once RRSI's rule is predicted to reject it used 57% of full re-evaluation's episodes but dropped a quarter of its acceptances; a post hoc diagnosis traces this to stops that treated an estimated cost change as known. It still passed a registered non-inferiority test on recorded decisions, as keeping the incumbent would. The independent-loop comparison did not establish non-inferiority at its preregistered five-point margin (-4.4 points, 95% interval -9.6 to $+0.7$). Under shared calibration, the estimated difference was -0.8 points (95% interval -3.4 to $+1.7$); this experiment did not preregister the same non-inferiority test. Its loops cost 10–12% less (not statistically established). These findings concern one loop, one acceptance rule and one stopping rule. Cheaper verification should be reported with its recall of the reference's acceptances, the held-out value of those it drops and the calibration of its stops.

1 Introduction

A growing family of systems improves LLM agents from their own failures. The agent runs, an analyst model reads the failed trajectories, a proposer changes the agent's *harness*, the prompts, tool descriptions and code around a frozen model, and the loop keeps the change or discards it (Shinn et al., 2023; Zhao et al., 2024; Hu et al., 2025; Zhang et al., 2025a; Agrawal et al., 2026; Chen et al., 2026; Zhang et al., 2026b; Xia et al., 2026). The check that decides is where these systems differ most: some replay the failure with the change applied (Ma et al., 2025; Lin et al., 2026; Shah, 2026), some ask an LLM judge, some re-run tasks that used to pass (Chen et al., 2026), and RRSI re-evaluates every candidate twice on its whole training set (Xia et al., 2026). What every check is meant to estimate is a change's *deployment gain*, how much it raises success on tasks the loop has not seen, and that gain can differ from what the loop observes on its own tasks: RRSI reports in-distribution gains that shrink out of distribution (Xia et al., 2026).

This paper asks two questions about verification evidence. *Which evidence predicts a change's deployment gain?* And *is the evidence worth what it costs?* Answering the first needs ground truth that verification does not provide: deploying the changes, from the first step, on independent tasks. Answering the second needs an account of what verification costs and what the change it selects is worth over a deployment horizon, which we write as $\mathrm{Net}(N)$ (§3). We price every piece of evidence in agent episodes and answer both questions in two studies, for one loop at a time: rather than rank kinds of evidence by how well they predict single changes, which test sets of this size cannot resolve (§4.1), we audit individual verification decisions against deployment. RRSI already treats generalization and constrained selection as central (Xia et al., 2026); we add an audit of each decision.

A preliminary study of one round of attribution-driven text patches on τ^2-bench (Barres et al., 2025) could not resolve differences between verification policies (Study 1, Appendix S2). **Study**

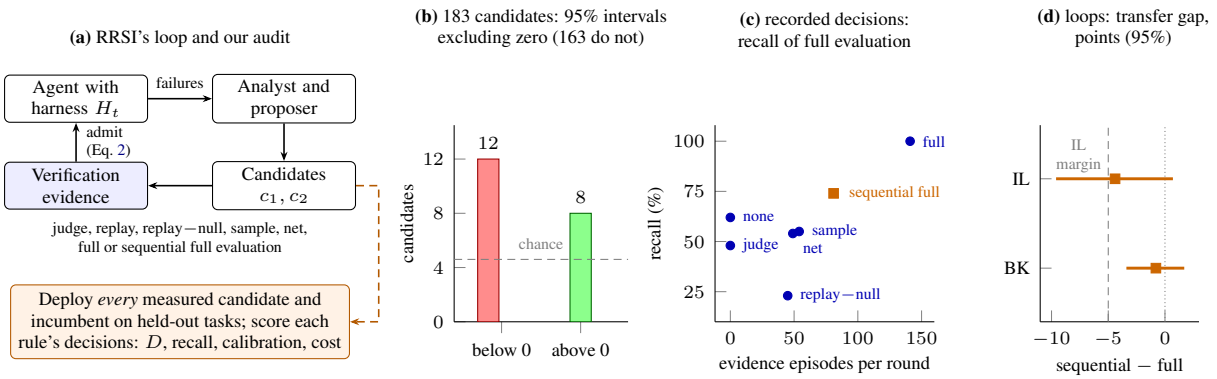


Figure 1: Overview. (a) RRSI's loop proposes harness changes and admits them by Eq. 2 on one kind of verification evidence; we deploy every measured candidate and incumbent on held-out tasks and score what each rule would have decided. (b) Unadjusted per-candidate 95% intervals of held-out gain (dashed: about 4.6 per side expected by chance). (c) Share of full evaluation's 21 acceptances on r2+r3 that each rule makes, against its evidence episodes per round (Table 2). (d) Sequential minus full evaluation in held-out transfer after ten airline rounds, for independently calibrated loops (IL) and loops sharing a calibration (BK), with 95% intervals; dashed: IL's registered non-inferiority margin, a test BK did not register.

2 (§4, registered) tests evidence with the power and the modification space Study 1 lacked. We run RRSI's open-source loop (Xia et al., 2026), which edits harness code over many rounds, score each kind of evidence in its selection step, and deploy every candidate and incumbent on held-out tasks (Figure 1). Few candidates measurably helped held out; evidence that does not run the changed agent on solved tasks would have accepted a change that crashed the harness and one that blocked legitimate actions; and no rule was shown to outperform keeping the incumbent or to pay for its verification. Sequential evaluation, which stops a candidate's evaluation once RRSI's rule is predicted to reject it, used 57% of full re-evaluation's episodes on the loop's recorded decisions but dropped a quarter of its acceptances, because its stops were overconfident where RRSI admits a change for being cheaper or new; it passed a registered non-inferiority test, as keeping the incumbent does. Six independent airline loops per arm driven by it did not establish non-inferiority to full evaluation's at the registered five-point margin: they gained 4.4 points less held out (95% $[-9.6, +0.7]$), and their separately calibrated noise bands differed by chance. With one calibration shared by five loops in each of four blocks, sequential loops gained 0.8 points less (95% $[-3.4, +1.7]$; this experiment registered no non-inferiority test), for 37% fewer candidate-evaluation episodes but only 12% lower loop cost, which the proposer dominates. An ablation that separates admission from the proposer's history resolved neither effect: the admission effect, -1.6 points, does not survive adjustment for multiple contrasts.

Our contributions are an audit of individual verification decisions, a methodological case study rather than a ranking of evidence or a new loop: a candidate-level evaluation protocol that deploys every candidate, accepted or rejected, with an explicit cost account; a diagnosis of why decision values against a reference that gains little cannot reveal lost acceptances, and of the miscalibrated stops that lose them; registered loop comparisons that examine admission, the proposer's history, calibration and total cost; and the environments, banks and evidence of both studies. The loop, its regularised edits and its selection rule are RRSI's (Xia et al., 2026); we change only the evidence its selection step uses.

2 Related Work

Self-improving agents and how they verify. Reflexion, Self-Refine and ExpeL turn feedback into verbal lessons (Shinn et al., 2023; Madaan et al., 2023; Zhao et al., 2024); ProTeGi, OPRO, TextGrad and DSPy rewrite prompts (Pryzant et al., 2023; Yang et al., 2024; Yuksekgonul et al., 2025; Khattab et al., 2024); ADAS, the Darwin Gödel Machine and GEPA search over designs (Hu et al., 2025; Zhang et al., 2025a; Agrawal et al., 2026); ACE evolves a context playbook (Zhang et al.,

2026b). HarnessFix repairs harness code and promotes a bundle of repairs only if it solves more validation tasks while breaking at most two (Chen et al., 2026); Bayesian-Agent applies threshold rules to posteriors over skills (Wu et al., 2026). RRSI regularises harness self-improvement and selects by full re-evaluation against a calibrated noise floor (Xia et al., 2026). These systems report the success of the final agent; we evaluate the evidence behind each decision.

Attribution and verification by intervention. Locating the responsible agent and step is far from solved (Zhang et al., 2025b; Liu et al., 2026); attribution models, taxonomies and dependency tracing address it (Zhang et al., 2026a; Zhu et al., 2025; Li et al., 2026; Cemri et al., 2025, 2026). DoVer validates hypotheses by editing a message or plan and re-running (Ma et al., 2025); REFLECT tests patches by targeted replay (Lin et al., 2026); CausalFlow and Causal Agent Replay score step interventions causally (Bonagiri et al., 2026; Shah, 2026); AgentDebugX gates reruns by policy (Zhu et al., 2026). Closest to our question, a failure critic with an AUROC of 0.94 lowered one agent's success by 26 points when its predictions triggered interventions within runs, because interventions also derail runs that would have succeeded (Vasudev et al., 2026). We study the analogous gap for harness changes selected over many rounds, between a rule's recorded decisions and the loops it drives. Program repair has long warned that a patch passing the failing test may not fix the fault (Smith et al., 2015).

Sequential evaluation and racing. Hoeffding races and F-race discard configurations once a bound or a Friedman test separates them from the best (Maron and Moore, 1993; Birattari et al., 2002); successive halving and Hyperband spread a budget over rounds of elimination (Jamieson and Talwalkar, 2016; Li et al., 2018); fast cross-validation drops parameter settings by sequential tests across growing subsamples (Krueger et al., 2015); clinical trials stop for futility when the predictive probability of a conclusive result is small (Spiegelhalter et al., 1986); and online experiments use always-valid inference so that continuously monitored A/B tests keep their error rates (Johari et al., 2017). Our sequential rule is a futility stop whose target is a self-improvement loop's admission rule, with its noise band, cost budget and novelty branch.

3 Verification Evidence and Deployment Gain

Changes and their deployment gain. A loop holds an incumbent harness h and proposes a change c: a rewritten prompt section or tool description, a code edit, or a set of these. Its *deployment gain* $\Delta p(c)$ is the difference in success rate between h with c and h alone, measured from the first step on tasks independent of everything the loop saw. A verification rule spends C_{verify} episodes on evidence, forms an estimate $\widehat{\Delta}$ of the change's effect and accepts the change if the estimate clears a threshold.

Evidence. *None* accepts the proposed change. An *LLM judge* reads the change and the failures it targets and predicts its effect, at a cost in tokens but no episodes. A *targeted replay* resumes a failed trajectory from the failing step with the change and succeeds with probability $b + (1 - b)q$, where q is the share of such failures the change repairs; a *null replay* does the same without the change and measures the *spurious recovery* rate b, without which replay success cannot be interpreted. A *regression run* is a previously solved task, from the first step, with the change; a *sample* or *full evaluation* runs candidate and incumbent on a random sample of, or all, training tasks. Replays and judges see only the failures a change targets; only regression runs and evaluations see harm to tasks that already succeed.

Cost and value. Against a reference rule, a rule E whose choices change deployed success by Δp and running cost by Δc_{run} per episode, which spends C_{verify} on evidence and changes the one-off cost of reviewing the applied changes by ΔC_{change}, is worth, over the next N deployed episodes,

$$\text{Net}(N) = N\,(v\,\Delta p - \Delta c_{\text{run}}) - C_{\text{verify}} - \Delta C_{\text{change}}, \quad (1)$$

with costs in episodes of the unmodified agent and v the value of one success (Appendix S3). We count *episodes*, runs of the agent on a task (a replay from a failing step counts as one); *episode equivalents*, which price replays and checks by their tokens and are the unit of C_{verify}; and *dollars* at list prices, for a whole loop's cost, including the analyst's and proposer's model calls. For large N evidence pays only if the changes it selects are better or cheaper to run, so we measure the *decision value* of a rule that uses it, the deployment gain of its choice averaged over rounds, zero when it

keeps the incumbent; how well its estimates track single changes' gains is bounded by the test sets' resolution (§4.1).

4 Study 2: Verification Evidence Inside RRSI

Study 2 builds on RRSI (Xia et al., 2026), whose proposer edits harness code over many rounds of natural failures. We run RRSI's loop, collect every kind of evidence for every candidate it measures, deploy every candidate, accepted or rejected, and ask what each kind of evidence would have decided. Table 1 lists the experiments, and Appendix B describes the port, the registrations and every change made after them.

Loop. We vendor RRSI's code (Apache 2.0) and change only its model backend, its cost accounting and one rule for empty proposals. Analyst, proposer, critic and judge are DeepSeek-V4-Pro with thinking disabled; the agent and user simulator are DeepSeek-V4-Flash at temperature 0; the starting harness is τ^2's reference agent. Each round the proposer drafts $m = 2$ candidates under an edit budget that anneals from 4 to 1, a critic rejects edits that hard-code task data or touch the grader, and RRSI evaluates the candidates that pass a smoke test twice on every evolve task. With S' a candidate's evolve success, S^* the best so far, ΔS and ΔC its changes in success and relative running cost against the incumbent and ν its novelty, RRSI admits it if

$$S' \geq S^* - \delta \text{ and } \begin{cases} \Delta C \leq \beta_0 + \beta_1 \Delta S & \text{if } \Delta S > \delta, \\ w_s \Delta S - w_c \Delta C + w_n \nu > 0 & \text{otherwise,} \end{cases} \quad (2)$$

where the noise band δ is calibrated from three evaluations of the base harness and, with RRSI's weights ($w_s = 0$), only a cheaper or untried change passes inside the band; it rejects a harness that raises errors in more than 2% of trials, and the admissible candidate with the highest success wins (Appendix B). Evolve tasks are τ^2's training split (74 retail, 30 airline) and held-out tasks its test split (40 and 20). The loop ran $T = 20$ rounds per domain on three independent trajectories, r1–r3.

Evidence and decisions. For each measured candidate we collect the evidence of §3 as estimates of its change ΔS in evolve success (Table 4, Appendix B): an LLM judge's prediction from the diff and the failures it targets (no episodes); *sample*@b, the paired difference on b evolve episodes; *replay*@b, the failure share times the success rate of b targeted replays of failed evolve trials; *replay−null*@b, which subtracts the success rate of null replays with the incumbent; and *net*@b, which splits b between both kinds of replay and regression runs on solved tasks, with or without a Beta prior on the repair rate. Budgets of 10, 40 and 80 episodes per candidate are subsamples of one collection (Appendix B); a replay re-executes the recorded prefix through the new harness and goes live where it would first act differently. Every rule decides with RRSI's rule, with ΔS replaced by its estimate and δ by twice the estimate's standard error; full evaluation keeps RRSI's calibrated δ. Appendix S19 decides every rule with one criterion.

Sequential full evaluation. RRSI evaluates a candidate on N paired episodes, two per evolve task. The sequential rule runs them in batches of 10 in random order. After n episodes with mean paired difference m and running cost change $\hat{c}$, it predicts the final $\Delta S \sim \mathcal{N}(m, v_n)$ with $v_n = \left(\frac{N-n}{N}\right)^2 s^2 \left(\frac{1}{n} + \frac{1}{N-n}\right)$, where s^2 is the larger of the running variance of the paired differences and $S_t(1 - S_t)$, and computes the probability p_n that Eq. 2 will hold with $\Delta C = \hat{c}$ and the known novelty. It stops the candidate if $p_n < \gamma = 0.05$ or its harness errors already exceed 2% of N. The *cost-aware rule* (post hoc, Appendix S20) instead draws the final $\Delta C \sim \mathcal{N}(\hat{c}, v_n^c)$, with v_n^c of the same form from the per-pair token differences, independently of ΔS, and integrates Eq. 2 over both. A candidate that is not stopped completes its evaluation and is decided as under full evaluation. Stopping thus changes only admission: it can lose an acceptance but never add one. A stopped candidate enters RRSI's history with its early-stopped estimate. *Sequential sampling* caps N at 80 sampled episodes and decides with the sample's own band.

Ground truth and estimand. Every incumbent and measured candidate is deployed from the first step on the held-out tasks, 240 episodes each (retail 40 tasks × 6, airline 20 × 12), and the base and final harnesses on 400 and 240. A candidate's deployment gain $\Delta p(c \mid H_t)$ is its held-out success minus that of the incumbent H_t it was proposed against, paired by task. A rule E's *decision value* on a set $\mathcal{T}$ of rounds of both domains is $D(E) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \sum_c P_E(c \mid t)\, \Delta p(c \mid H_t)$, the

Experiment	Question	Independent units	Endpoint
r1 (R)	Which evidence decides best inside RRSI?	1 loop per domain, 40 rounds in 7 incumbent blocks; 40 and 20 test tasks	(P1) transfer of the final harness; (P2) D at $b = 40$
r2+r3 (R, primary)	Are cheaper rules non-inferior to full evaluation?	2 loops per domain, 80 rounds in 23 blocks; same tasks	D, non-inferiority at 0.75 points per round, Holm
r4 (D); E3 (R)	Does a thinking proposer, or a code agent, change this?	1 airline loop, 10 rounds; 1 AppWorld loop, 20 rounds, 168 test tasks	as r2+r3
CL (D); IL (R)	Do loops driven by sequential evaluation transfer as well?	6 + 6 airline loops (4 + 4 concurrent), 10 rounds; 20 test tasks	transfer at round 10, non-inferiority at 5 points; cost
BK (R)	Does the loss replicate under shared calibration? Admission or history?	4 blocks of 5 airline loops sharing a calibration, 10 rounds; 20 test tasks	transfer at round 10: rule differences, admission and history effects
Post hoc	Where does the sequential rule err?	as above	recall, calibration, cost-aware rule, noise bands, other sensitivity analyses

Table 1: The experiments. R: registered; D: registered as descriptive; Study 1 (one round of text patches, Appendix S2) is not shown. Rounds sharing an incumbent form a block, the unit of the block bootstrap; trials within units are repetitions (Table S35). D: decision value, the mean held-out gain of a rule's choice per round. Appendix A lists the terms.

mean expected deployment gain of its choice per round (zero when it keeps H_t), with P_E taken over 400 subsamples of the evidence. For r1 the registered endpoints are (P1) *transfer*, the held-out gain of the final harness over the base, and (P2) D at $b = 40$ with three Holm-adjusted contrasts (Appendix S17).

Further trajectories. Because r1's values rested on one trajectory per domain and its registered analysis gave RRSI's error check to full evaluation only, we registered r2 and r3 before any of their episodes ran, with this study's primary analysis: D over their 80 rounds with the error check applied to every rule's own episodes (the *common error check*), and one-sided non-inferiority tests of sample@40, net@40 and replay−null@40 against full evaluation at a margin of 0.75 points per round, Holm-adjusted at $\alpha = 0.05$; the two sequential rules form a second family. A bootstrap resamples blocks of rounds that share an incumbent and, within each trajectory and domain, one draw of held-out tasks shared by all its candidates, incumbents and rules, so its intervals generalise to new held-out tasks and to rounds like the observed ones, not to new improvement runs, which only the loop comparisons below resample; brackets give 90% intervals, the registered level, unless marked 95%. E3 runs the same loop and analysis on AppWorld (Trivedi et al., 2024) (Appendix S18).

4.1 Results

The loop. Over r1–r3 RRSI drafted 240 candidates and measured 154, 106 of which changed harness code. It accepted 5 candidates on r1, all in the first five rounds, and 21 on r2 and r3, as late as round 19; 19 of the 26 were within the noise band. Candidates' mean deployment gain was -1.5, -1.25 and $+0.14$ points on r1–r3. 63 of 154 had positive point estimates, but held-out tasks resolve a single candidate's gain only beyond about four points: unadjusted 95% intervals lay above zero for 8 and below for 10 (on AppWorld for none and 2 of 29; post hoc, Appendix S17), against 4–5 per side expected by chance.

Transfer (P1). Airline's evolve gains transferred on r1 and r2, by $+8.75$ $[+2.5, +15.8]$ and $+7.9$ $[+2.5, +14.2]$ points against $+13.3$ on evolve tasks. On r3 its six acceptances, all within the band, left evolve success unchanged ($+1.7$ $[-2.5, +5.8]$ held out). Retail's evolve gains of 4.1, 4.7 and 7.4 points never transferred: -0.25 $[-3.0, +2.75]$, 0.0 $[-2.0, +1.75]$ and -4.25 $[-7.5, -1.0]$.

First trajectory. On r1, one candidate (retail, round 6) added a check that reads an attribute the harness never sets: the harness raised errors in half of its evolve episodes and held-out success fell by 51.7 points, yet the critic passed it and the judge predicted a gain. It causes most of the losses of no verification, the judge and replay without null replays. Full evaluation's decision value on r1 was $+0.22$ points per round $[-0.06, +0.62]$ (Ap-

Rule	Ep.	D [90%]	$p_{0.75}$	$p^{\text{sh}}_{0.75}$	Agree	Recall
Full (RRSI)	141	+0.04 [−0.35, +0.43]	–	–	1.00	1.00
Keep incumbent	0	0	0.004†	0.003†	0.74	0.00
None	0	−0.47 [−1.52, +0.62]	–	–	0.25	0.62
LLM judge	0	−0.40 [−1.41, +0.67]	–	–	0.30	0.48
Sample@40	54	−0.06 [−0.58, +0.45]	0.045	0.056	0.67	0.55
Net@40	49	−0.10 [−0.62, +0.39]	0.045	0.056	0.63	0.54
Replay−null@40	45	−0.39 [−0.79, −0.03]	0.14	0.14	0.68	0.23
Seq. full	81	+0.02 [−0.34, +0.35]	<0.001	<0.001	0.93	0.74
Seq. sample	53	+0.07 [−0.26, +0.41]	<0.001	<0.001	0.78	0.48

Table 2: Primary analysis, r2+r3 (80 rounds; from Tables 5 and 6). Ep.: evidence episodes per round; D in points per round; $p_{0.75}$: non-inferiority to full evaluation at the registered margin, Holm-adjusted (†post hoc); $p^{\text{sh}}_{0.75}$: the same with one task draw shared by r2 and r3 (post hoc); Agree: share of rounds with full evaluation's choice; Recall: share of its 21 acceptances the rule makes.

pendix S17).

Primary analysis (r2 and r3). On their 80 rounds full evaluation's decision value was only $+0.04$ points per round $[-0.35, +0.43]$ at 141 episodes (Table 2; every rule in Table 5), and no rule was shown to outperform keeping the incumbent. Replay−null@40 lost value, -0.39 $[-0.79, -0.03]$, 40% of it from one candidate (r3 retail, round 11) that held every write whose arguments it could not trace to earlier tool results: it raised no harness errors but blocked legitimate writes, lowering held-out success by 12.5 points; full evaluation, sample and net, which run solved tasks, rejected it, while replay−null and the judge accepted it. The registered non-inferiority tests against full evaluation, at a margin of 0.75 points per round, say little here, as keeping the incumbent passes too ($p = 0.004$; Table S22). Sample@40 and net@40 passed narrowly (Holm $p = 0.045$ each) and replay−null@40 did not ($p = 0.14$); sharing one task draw between r2 and r3, which use the same test tasks, gives the @40 rules 0.056 (post hoc), so their pass is borderline, sensitive to the dependence structure and to Monte Carlo error (Appendix S19). A rule can pass them while dropping many of the reference's acceptances.

Sequential evaluation on the recorded rounds. Sequential full evaluation, at 81 episodes per round (57%), made full evaluation's choice in 93% of rounds but only 74% [67, 81] of its 21 acceptances (Table 2). It passed the registered test, with a difference from full evaluation of -0.02 $[-0.12, +0.07]$ points per round ($p < 0.001$; sequential sampling $+0.03$ $[-0.26, +0.35]$ at 53 episodes), and post hoc a margin of 0.25 points, which keeping the incumbent fails, with any one trajectory and domain left out (Table S34); but the acceptances it would drop gained no less held out ($+0.30$ against $+0.11$ points), almost all in success rather than running cost; pooled, it lost little because full evaluation's acceptances were worth little (Table 6, Appendix S21). Their value was concentrated: full evaluation gained $+0.58$ points per round on airline rounds 0–9 and -0.19 to -0.10 elsewhere; there the rule's difference was -0.01 $[-0.17, +0.19]$ (the 2.4 acceptances it dropped were worth $+0.01$ points), and -0.10 $[-0.36, +0.09]$ with r1, point estimates of 0.1 and 1.0 points over ten rounds (Table S36). Simpler halvings with RRSI's own rule are less faithful: one trial per task keeps 63% of the acceptances, both trials on half the tasks 53% (Appendix S20).

Is the stopping rule calibrated? No (post hoc; Table 7, Appendix S20). Candidates stopped at a mean predicted probability of 0.02 would have been admitted by full evaluation in 8% of cases on r2+r3 and 13% in the six live loops below, 5.6 [2.5, 11.1] and 11.3 [3.0, 25.8] points more than predicted (95%, resampling incumbent blocks); AppWorld shows no detectable miscalibration (3%). Holding the cost change at its running estimate causes this: with the complete evaluation's cost change in the same formula, looks below γ were followed by admission in at most 2% of cases. The rule stopped about a third of the candidates RRSI admits inside its band for being cheaper or new, in recorded and live rounds alike, against 4% of those admitted above the band. The cost-aware rule, designed post hoc on these rounds, halves the stops inside the band and keeps 88% of full evaluation's acceptances on r2+r3 at 64% of the episodes (Table S29). Offline, the registered rule's recall on airline falls from 95% at a band of 2.5 points to 51% at 17.5, while the cost-aware rule keeps 87–96% (Table S28).

Independent loops. The analyses above score decisions in full evaluation's states. To compare loops, we registered two airline loops driven by sequential full evaluation (CL: cl1, cl2), then eight concurrent loops, four per selection step (IL; Appendix S22). With r2 and r3, six loops per arm ran rounds 0–9. Transfer after ten rounds averaged $+4.2$ points with sequential and $+8.7$ with full evaluation, a difference of -4.4 $[-9.4, -0.1]$ (Figure 1d); non-inferiority at the registered mar-

Arm	Admission	History	B1	B2	B3	B4	Mean	Acc.	Cand. ep.	Cost ($)
Full	full	full	$+6.7$	$+9.6$	$+11.2$	$+7.5$	$+8.8$	12	855	3.40
History only	full	early	$+4.6$	$+10.4$	$+14.2$	$+10.8$	$+10.0$	12	900 ($+5\%$)	3.60 ($+6.0\%$)
Cost-aware seq.	cost-aware	early	$+7.1$	$+8.3$	$+13.3$	$+10.0$	$+9.7$	11	685 (-20%)	3.31 (-2.5%)
Admission only	seq.	full	$+7.1$	$+5.0$	$+11.2$	$+7.5$	$+7.7$	11	900 ($+5\%$)	3.49 ($+2.7\%$)
Sequential	seq.	early	$+4.6$	$+8.3$	$+10.8$	$+7.9$	$+7.9$	10	535 (-37%)	3.00 (-11.8%)

Contrast	90% / 95% interval	Est.	Holm p (one-sided)
Sequential − full	$[-2.9, +1.2]$ / $[-3.4, +1.7]$	-0.8	0.68
Cost-aware − full	$[-1.1, +3.0]$ / $[-1.6, +3.5]$	$+0.9$	1.00
Cost-aware − sequential	$[-0.2, +3.8]$ / $[-0.8, +4.3]$	$+1.8$	0.28
Admission (main effect)	$[-3.0, -0.2]$ / $[-3.4, +0.2]$	-1.6	0.19
History (main effect)	$[-2.2, +3.7]$ / $[-4.2, +5.6]$	$+0.7$	1.00
Interaction	$[-3.9, +1.8]$ / $[-4.6, +2.5]$	-1.0	–

Table 3: Experiment BK (registered): transfer after ten rounds on airline (held-out success after round 9 minus the block's base, points, 20 tasks × 12 trials) of five loops in each of four blocks that share one calibrated noise band. Admission: the rule that decides acceptance; history: the estimates stopped candidates enter the proposer's history with. Acc.: changes accepted over the four blocks; Cand. ep. and Cost: candidate-evaluation episodes and dollar cost per loop, as if run alone, and their change against full evaluation. Contrasts: block means from the registered lineage model (t, Satterthwaite df) with the registered unadjusted 90% and post hoc 95% intervals, and in the last three columns a post hoc Holm p over the five registered contrasts, one-sided in each registered reading's direction; main effects $(s + a - f - h)/2$ (admission) and $(s + h - f - a)/2$ (history).

gin of 5 points was not established ($p = 0.38$), and the 95% interval, $[-9.6, +0.7]$, includes zero (Appendix D). The recorded decisions of airline rounds 0–9 of r1–r3 predict -1.0 $[-3.7, +0.8]$ (Table S39). But each loop calibrated its own noise band, and by chance the sequential loops' bands were wider (9.8 against 6.7 points); a wider band leaves more candidates to RRSI's in-band branch, where the rule's stops are least reliable, and adjusting for it post hoc halves the pooled difference, to -2.1 $[-6.2, +1.9]$, though the concurrent loops keep -5.8 $[-9.4, -2.2]$.

Blocked loops (BK). To separate the rule from its calibration and locate the loss, we registered a third comparison after IL's results (Appendix S23). Four calibration blocks each gave one noise band to five airline loops that differ only in selection: full evaluation (f), sequential full evaluation (s), the cost-aware rule (c) and two ablations. Loop h admits as f does but enters stopped candidates into the proposer's history with their early-stopped estimates, as s does; loop a admits as s does but shows the proposer every complete evaluation. Loops of a block share computation only while their states are identical, so each has the distribution of an independent loop of its rule; the four blocks, not the twenty loops, are the independent units. Sequential loops gained 0.8 points less than full evaluation's $[-2.9, +1.2]$ (Table 3), close to the recorded decisions' prediction: IL's loss did not replicate under shared calibration. A loss of about three points is not excluded, but one of five, IL's margin, is, by the 95% interval $[-3.4, +1.7]$ and the Bonferroni interval $[-4.1, +2.4]$ (Appendix S23); BK registered no non-inferiority test, so this reading is post hoc (pooled with IL and adjusted for the band, -1.7 $[-4.1, +0.8]$). The admission main effect, -1.6 $[-3.0, -0.2]$, met its registered reading, which uses unadjusted 90% intervals; at 95% its interval includes zero and Holm's adjustment over the five registered contrasts gives $p = 0.19$, so it is consistent with, but does not establish, a cost of the dropped admissions; without any one block it lies between -1.0 and -2.2 (Appendix S23). Four blocks had 80% power only for admission effects of 2.0 points, and a one-point loop difference needs 28 (post hoc). In 7 of the 16 rounds in which full evaluation would have accepted in s's states, the rule had stopped its choice. The history effect, $+0.7$ $[-2.2, +3.7]$ on 1.4 df, is unresolved. The cost-aware rule made full evaluation's choice in all 40 of its rounds at 80% of the candidate-evaluation episodes; its gains over the registered rule, $+1.8$ $[-0.2, +3.8]$, and over full evaluation, $+0.9$ $[-1.1, +3.0]$, are not resolved.

Cost. What sequential evaluation saves is candidate-evaluation episodes, not the loop's cost: its IL loops ran 39% fewer such episodes at 10% lower cost ($-\$0.35$ $[-0.57, -0.10]$; at 95% the interval includes zero), and its BK loops 37% fewer at 12% lower cost (descriptive; Table 3), because the proposal side's model calls dominate; the cost-

aware rule saved 20% of the episodes and 2.5% of the cost. Spent on 1.4 more rounds, the registered rule's saving would move its offline ten-round loss on airline only from -1.0 to -0.9 $[-3.2, +0.9]$ (post hoc; Appendix S20), a comparison that keeps full evaluation's states; no loops ran at matched total cost. Against keeping the incumbent, no rule's verification was shown to pay on r2 and r3: $\mathrm{Net}(10^4)$ at $v = 10$ was -108 episodes $[-500, +274]$ for full evaluation and $+10$ $[-315, +350]$ at best, for sequential sampling.

Other trajectories. Pooled over r1–r3, full evaluation gained $+0.10$ $[-0.18, +0.41]$ per round, and trajectories differed more than rules. On AppWorld (E3) the evolve gain of 4.2 points did not transfer (0.0 $[-2.1, +2.1]$); sequential full evaluation passed non-inferiority at 62% of the episodes and the @40 rules did not (Table S21).

5 Discussion

What the evidence shows. Our results identify failure modes of verification, not a ranking of evidence by predictive quality. Few changes measurably helped held out, so verification earned its place by rejecting harmful ones, and only evidence that runs the changed agent on solved tasks rejected both kinds we saw. Sorted by evidence (Appendix D), only three conclusions rest on registered analyses with adjusted inference: no rule was shown to outperform keeping the incumbent; the sequential rules were non-inferior to full evaluation on recorded decisions, as keeping the incumbent was; and sequential loops ran fewer candidate-evaluation episodes. The harmful and the dropped acceptances are registered but descriptive, the stops' miscalibration is a post hoc diagnosis whose fix BK checked descriptively, and the loop losses, their mechanism and the dollar saving are not established. The most secure contributions are thus the audit method and the localisation of one stopping rule's decision distortion; how much such distortion costs a loop's final agent, and which verification is best at a fixed budget, remain open.

Recorded decisions and loops. Scoring each decision once, in the states full evaluation chose, ignores that an accepted change becomes the base of later rounds and that every evaluated candidate enters the proposer's history; BK's estimate of an adverse effect of the first did not survive adjustment, and the second is unresolved. A calibration imbalance may have contributed to IL's loss, but the experiments do not isolate it; loops compared with each other should share or balance their calibration.

What recall adds. Recall detected this rule's problem, which its decision value could not: it dropped a quarter of full evaluation's acceptances, in recorded rounds as in the loops, but they were worth little in the primary analysis. Recall measures fidelity to the reference, not deployment quality (the evolve cost savings for which RRSI admitted many changes did not carry to held-out tasks; Appendix S21), but it adds what a decision value averaged over rounds in which the reference gains little averages away: whether a rule keeps the changes a loop then builds on. It should be reported with the held-out value of the dropped acceptances and the calibration of the stops on each branch of the acceptance rule; here they flagged the rule's mechanism before any loop ran, but whether higher recall predicts a better final agent has been checked on only two rules in one domain.

Implications for self-improvement loops. Loops that report only the final agent's held-out success (Chen et al., 2026; Zhang et al., 2026b; Xia et al., 2026) measure all their decisions at once; on retail and AppWorld, evolve gains of 4–7 points never transferred. Replays need null replays and regression runs. A stopping rule must model the uncertainty of every condition in the acceptance rule it predicts; without RRSI's cost and novelty branch the registered rule makes 90–100% of the far fewer acceptances (Appendix S20). Because proposal calls dominate a loop's cost, cheaper proposals, which we did not vary, are likely a larger lever than cheaper verification.

6 Conclusion

Auditing RRSI's decisions one candidate at a time, we found that few changes measurably helped held out, that evidence that never runs the changed agent on solved tasks would have accepted harmful ones, and that no rule's verification was shown to pay against keeping the incumbent. A sequential rule stayed close to full re-evaluation's decision value on recorded rounds but dropped a quarter of its acceptances; the cost of this to a loop's final agent, and the best verification at a fixed budget, remain open. Cheaper verification should report its recall of the reference's acceptances and its calibration.

Limitations

Models and domains. Both studies use one model family, with DeepSeek-V4-Flash as agent and DeepSeek-V4-Pro as analyst, proposer and judge, and two customer-service domains of τ^2-bench; Study 1 adds ShopDesk and Study 2 one AppWorld trajectory. Study 1's agent used the provider's default sampling and Study 2's uses temperature 0. Study 2's proposer is likely weaker than the model RRSI was developed with; a fourth trajectory with a thinking proposer was cut short because its drafts took over an hour and mostly ended without a proposal; on airline all six of its measured candidates helped, which suggests that a stronger proposer leaves less for verification to decide, but rests on one trajectory (Appendix B). Token prices for the models we used change; we report costs in episodes, and weighting tokens by list prices changes no ranking.

Study 1. Study 1's patches rewrite only policy sections, tool descriptions and a step limit; it runs a single round, mostly on injected faults (natural failures are 16–25% of training failures), and deploys patch sets only at budget 10, on 120 (retail) and 60 (airline) held-out episodes, where most of retail's gain comes from fault-injected episodes. Its bank analyses rest on a harm model fitted to at most 20 regression runs per patch, and its edge priors use an analyst accuracy of 0.7 unless stated.

Test tasks and resolution. Study 2's held-out sets are τ^2's test splits of 40 and 20 tasks, on which most candidates' gains were not resolved: of the 183 measured candidates, unadjusted per-candidate 95% intervals lay above zero for 8, about what chance alone would give, and below zero for 12. Every airline loop (r1–r4, CL, IL and BK) and most post hoc analyses use the same 20 held-out tasks, and several analyses (the noise-band adjustment, the cost-aware rule, the noise-band replays) were designed after held-out results of earlier experiments on those tasks were known; task bootstraps capture sampling error on these tasks, not the effect of analysing them repeatedly, and τ^2's airline domain has no further tasks to hold out. Registering an experiment before its episodes run does not make these tasks a fresh test set. Resolution limited these studies more than price: a round of full evaluation costs about 25 cents of τ^2 compute, but comparing evidence needs per-round differences of about 0.7 points (Appendix S17), and more trajectories and held-out tasks would help more than more trials (Table S35). Three trajectories per domain show that between-trajectory variation is as large as the differences between rules, and the pooled analysis treats trajectories as fixed. The non-inferiority margin of 0.75 points per round was set before any result, but it is large relative to the decision values we observed.

Recorded decisions and loops. Study 2's decision values are conditional on RRSI's trajectories: they score each rule's decisions in the states RRSI visited. The loop comparisons use one domain and ten rounds, at equal rounds rather than equal cost; the equal-cost comparison of Appendix S20 is offline and assumes extra rounds are worth what full evaluation's later rounds were. IL's arms mix loops run at different times (cl1, cl2, r2 and r3, whose outcomes were known when it was registered, and eight concurrent loops), with arm labels fixed rather than randomised and noise bands that came out wider in the sequential arm. Its registered 90% intervals were not adjusted for its three primary outcomes; with 95% intervals and Holm's adjustment, neither its transfer difference nor its cost difference is established (Appendix D). BK has four blocks, its independent units, which give 80% power only for admission effects of about two points; its bands (5.4 to 8.2 points) were narrower than those of IL's sequential loops, and a wider band may enlarge the rule's loss. Its registration read each of five contrasts by an unadjusted 90% interval; the admission effect does not survive a 95% interval or Holm's adjustment, its history effect is imprecise (1.4 Satterthwaite df), its loops share computation while their states are identical, and its cost-aware loops' gain over the registered rule was not resolved. Whether calibration explains the difference between IL and BK is not identified: the two experiments also differ in their trajectories and band ranges.

Scope of the diagnosis. The stops' miscalibration appeared on τ^2's recorded rounds and in the live loops but not detectably on AppWorld, and only airline's loops improved held out, so what it costs a loop has been measured in one domain. Our diagnosis concerns RRSI's acceptance rule, under which 19 of the 26 acceptances of r1–r3 fell inside the noise band, where cost and novelty decide, and one stopping rule; it does not establish general properties of replay, judging or sequential evaluation, and a loop with a different acceptance criterion may

behave differently. The cost-aware rule's fidelity in loops rests on 11 acceptances in bands of 5.4 to 8.2 points; offline it held for bands of 2.5 to 17.5 points, on recorded trajectories only. $\mathrm{Net}(N)$ sets the change cost to zero; charging each accepted change a fixed review cost favours rules that accept less and leaves the order of the sequential rules and full evaluation unchanged (Appendix S19). No rule's verification was shown to pay against keeping the incumbent on r2 and r3, so our cost comparisons rank rules that may all cost more than they return in this loop.

Registration and post hoc analyses. The analysis of other noise bands is post hoc. BK was registered after IL's results, and the cost-aware rule was designed post hoc on the recorded rounds before it was registered for BK. Post hoc are, on recorded decisions, the smaller margins, the agreement rates and their split by full evaluation's decision, the decision-rule ablation, the coverage, calibration, threshold, matched-cost and leave-one-out analyses, the simpler halvings, the noise-band replays, the re-decision without the cost and novelty branch, the split by domain and stage, the utility of the dropped acceptances, the clustered calibration intervals, the per-candidate resolution, the task draw shared across trajectories and the offline evaluation of the cost-aware rule; and, on loops, the matched offline comparison, the noise-band adjustment, the accounting of IL's loops, BK's sensitivity, leave-one-block-out and power analyses and the equal-cost comparison (Appendix D sorts the conclusions by the evidence behind them). On r1, the registered analysis gave RRSI's error check to full evaluation only, so the comparison with a common check is post hoc there; its held-out count was raised after six rounds, when some held-out episodes had run and a few results had been seen (Appendix B), and the registered count gives the same conclusions. While r2 and r3 ran, a few of their held-out success rates appeared in progress logs before analysis; the analysis code had been fixed and tested on r1 only. The sequential rules were designed on r1's rounds and confirmed on r2 and r3. r1's component labels did not recognise our harness's files, which may have changed what its proposer was told to explore; r2 and r3 use corrected labels, so the trajectories differ in that respect as well as in sampling.

Ethics Statement

Loops of the kind we study modify software that is then deployed. In Study 2 few proposed changes measurably raised held-out success, and two harmful ones, one that crashed the harness in half of its episodes and one that blocked legitimate writes, would have been accepted by checks that did not run previously solved tasks. That Study 1 detected no gain from verification does not show that verification is unnecessary; our measurements cannot exclude small harms per change and should not be read as licence to deploy changes without review. On ShopDesk a single patch silently broke a quarter of tasks (Appendix S15). Our environments use synthetic customer data only.

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A Terms and Supplementary Appendices

The supplementary material contains the complete appendices, numbered S1–S26: Study 1, its accounting framework, verification policies and ShopDesk (Appendices S2–S15); Study 2's port and registrations in full, the first trajectory, AppWorld, margins and decision rules, the validation of the stopping rule, the split by domain and stage, the independent loops and BK (Appendices S16–S23); and the multi-fidelity policy (Appendix S26). The appendices below repeat what the body relies on most.

Terms and Abbreviations

Harness, change, candidate. The prompts, tool descriptions and code around the frozen agent model; a proposed edit to it; a change the loop measures.

Incumbent. The harness a round's candidates are proposed against.

Deployment gain, transfer. A candidate's held-out success minus its incumbent's; a loop's held-out success after round 9 minus that of the base harness.

Evolve and held-out tasks. τ^2's training split, on which the loop evaluates candidates, and its test split, on which we deploy them.

Full evaluation. RRSI's selection: every candidate is run twice on every evolve task.

Noise band δ. RRSI's threshold, calibrated from three evaluations of the base harness. A candidate above it is admitted if its cost change is within budget; inside it (the *in-band branch*), if a score of its success change, cost change and novelty is positive.

Sequential full evaluation (registered rule). Full evaluation run in batches of 10 that stops a candidate once the predicted probability that RRSI's rule admits it falls below $\gamma = 0.05$, with the cost change held at its running estimate.

Cost-aware rule. The same, with the uncertainty of the final cost change modelled (Appendix S20).

Admission, history. Which candidates a rule can accept; the scores of evaluated candidates that RRSI shows the proposer in later rounds.

Decision value D. The held-out gain of a rule's choice, averaged over rounds, zero when it keeps the incumbent.

Agreement, recall. The share of rounds in which a rule makes full evaluation's choice; the share of full evaluation's acceptances it makes.

r1–r4, E3. Recorded trajectories of RRSI driven by full evaluation on τ^2 (r4 with a thinking proposer) and on AppWorld.

CL, IL, BK. Loops driven by sequential evaluation (two, descriptive); the registered comparison of six independent loops per rule; the registered blocked comparison of five rules in four blocks that share one calibration (Appendix S23).

Episode, episode equivalent. One run of the agent on a task; the token cost of a replay or check relative to an episode.

B Study 2: Port, Evidence and Registrations

Port. RRSI's loop is vendored from its public repository (Apache 2.0) with the upstream copyright headers kept. We changed the model backend (DeepSeek's Anthropic-compatible endpoint, thinking disabled, token usage recorded per role) and one rule: a proposal that ends with no file change and no declared edit is returned to the proposer, at most three times, instead of ending the round with no candidate. RRSI's selection rule (Eq. 2) is unchanged. Our τ^2 adapter implements RRSI's domain interface with a frozen episode driver and a starting harness with four hooks: choose the next action, check a tool call before it runs, render a tool result, and rebuild per-episode state from a recorded prefix. At temperature 0 our episode loop matches τ^2's own orchestrator (retail 90.0% against 90.8%, airline 71.7% against 71.7%). Besides RRSI's critic, a pattern check rejects added lines that contain task identifiers, refer to the grader or to τ^2's task data, import the benchmark or write files.

Evidence. For every evolve task the incumbent failed (at most 20 per round), replays start at the earliest step the round's failure digest cites and run 4 times with each measured candidate and 4 times with the incumbent (null replays); sample and regression pairs are drawn from the full evaluations. Budgets of 10, 40 and 80 episodes are subsamples of this one collection, drawn 400 times per round. Each estimate goes through RRSI's rule with twice its standard error as noise band and its measured token cost.

Evidence	Episodes	Estimate of ΔS
None	0	accept a candidate (at random if two)
LLM judge	0	predicted fixes − predicted breaks
Sample@b	b	paired difference on b evolve episodes
Replay@b	b	failure share × repair rate
Replay−null@b	b	failure share × (repair − null rate)
Net@b	b	replay−null − success share × regressions
Net@b, prior	b	net@b with a Beta prior on the repair rate
Full (RRSI)	148 / 60	paired difference on all evolve episodes

Table 4: Evidence compared in Study 2, per candidate, with $b \in \{10, 40, 80\}$; full evaluation costs 148 (retail) or 60 (airline) episodes. The priors have means 0.41 or 0.29 (audited analyst accuracy), 0.7 and 0.2. Every rule decides with RRSI's selection rule.

Registrations and changes after them. Every experiment was registered before any of its episodes ran (`verify/PREREGISTRATION*.md`): r1; r2 and r3, with the sequential rules added as an addendum before any of their decision values was computed; E3, CL, IL and BK; and r4 as descriptive. On r1 four changes were logged: before any held-out episode, a bug fix in the failure share and the implementation of $\mathrm{Net}(N)$; after six rounds, when some held-out episodes had run and a progress log had shown a few success rates, held-out trials per harness were raised from 160 to 240, and the registered 160 give the same conclusions. While r2 and r3 ran, a malformed reply to RRSI's digester was fixed, the container restarted twice and every run resumed from its last settled step; progress logs showed the held-out success of four rounds' harnesses, from which nothing was decided. CL and BK resumed after restarts in the same way, and no held-out result of BK was examined before all its deployments had finished. r4, with a thinking proposer, was stopped after two retail rounds and ran ten airline rounds, in which 14 of 20 drafts ended without a proposal. Appendix S16 gives the full account, the estimators and the costs.

C Further Results

Table 5 gives every rule's decision value, Table 6 splits each rule's difference from full evaluation by full evaluation's acceptances and Table 7 checks the stopping rule's calibration. Appendices S17–S23 give the remaining analyses.

D Conclusions by Evidence

Table 8 sorts the main conclusions by the analyses behind them and says which would remain with only prospectively specified analyses and inference

		Decision value D [90%], common error check			r2+r3			
Evidence	b	r1 (40 rounds)	r2+r3 (80, primary)	r1–r3 (120)	Accepts	Agree	Ep.	$\mathrm{Net}(10^4)$
Keep incumbent	0	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	0.00	0.74	0	0
None	0	−1.87 [−3.52, +1.53]	−0.47 [−1.52, +0.62]	−0.94 [−1.94, +0.22]	0.91	0.25	0	−818
LLM judge	0	−1.72 [−3.30, +1.37]	−0.40 [−1.41, +0.67]	−0.84 [−1.75, +0.27]	0.81	0.30	0	−731
None + check	10	−0.58 [−1.88, +1.56]	−0.47 [−1.52, +0.62]	−0.51 [−1.39, +0.46]	0.91	0.25	13	−831
Judge + check	10	−0.43 [−1.53, +1.41]	−0.40 [−1.41, +0.67]	−0.41 [−1.20, +0.47]	0.81	0.30	13	−744
Full (RRSI)	all	**+0.22** [−0.04, +1.11]	+0.04 [−0.35, +0.43]	**+0.10** [−0.18, +0.41]	0.26	1.00	141	−108
Sequential full	≤all	+0.15 [−0.03, +0.79]	+0.02 [−0.34, +0.35]	+0.06 [−0.18, +0.32]	0.20	0.93	81	−71
Sequential sample	≤80	+0.12 [−0.15, +0.98]	**+0.07** [−0.26, +0.41]	+0.09 [−0.15, +0.38]	0.23	0.78	53	10
Sample	10	−0.04 [−0.42, +0.77]	−0.08 [−0.54, +0.37]	−0.07 [−0.39, +0.30]	0.39	0.60	13	−107
Replay	10	−0.37 [−1.26, +0.88]	−0.67 [−1.56, +0.17]	−0.57 [−1.20, +0.13]	0.63	0.44	13	−1,000
Replay−null	10	−0.05 [−0.25, +0.16]	−0.36 [−0.74, −0.01]	−0.25 [−0.53, −0.00]	0.18	0.69	11	−594
Net	10	−0.07 [−0.54, +0.93]	−0.17 [−0.73, +0.35]	−0.13 [−0.52, +0.30]	0.48	0.56	12	−217
Sample	40	+0.04 [−0.35, +0.93]	−0.06 [−0.58, +0.45]	−0.02 [−0.37, +0.38]	0.40	0.67	54	−119
Replay	40	−0.40 [−1.56, +1.34]	−0.74 [−1.83, +0.36]	−0.63 [−1.46, +0.30]	0.86	0.33	50	−1,126
Replay−null	40	−0.07 [−0.34, +0.21]	−0.39 [−0.79, −0.03]	−0.29 [−0.58, −0.01]	0.20	0.68	45	−674
Net	40	−0.04 [−0.53, +0.76]	−0.10 [−0.62, +0.39]	−0.08 [−0.45, +0.31]	0.44	0.63	49	−152
Sample	80	+0.08 [−0.30, +1.06]	+0.04 [−0.46, +0.57]	+0.06 [−0.30, +0.47]	0.38	0.78	94	−57
Replay	80	−0.40 [−1.55, +1.34]	−0.71 [−1.80, +0.42]	−0.60 [−1.45, +0.33]	0.88	0.31	56	−1,101
Replay−null	80	−0.12 [−0.60, +0.45]	−0.51 [−1.03, −0.05]	−0.38 [−0.74, −0.02]	0.25	0.65	84	−848
Net	80	−0.02 [−0.53, +0.81]	−0.04 [−0.55, +0.45]	−0.03 [−0.42, +0.36]	0.41	0.69	98	−123

Table 5: Study 2: decision value D of each rule with the error check applied to every rule's own episodes, the held-out gain of its choice over the incumbent in points per round (keeping the incumbent scores 0), with 90% intervals from a bootstrap of blocks of rounds that share an incumbent and of held-out tasks, within domain and trajectory. r1: the first trajectory per domain (its registered analysis is in Table S20); r2+r3: the registered primary analysis; r1–r3: pooled. + *check* rules run 10 episodes of each candidate only for the check; sequential rules stop a candidate's episodes early. Accepts (mean probability of accepting a candidate in a round), Agree (share of rounds in which the rule makes full evaluation's choice), evidence episodes per round, and $\mathrm{Net}(10^4)$ (Eq. 1 against keeping the incumbent, $N = 10^4$ deployed episodes, a success worth $v = 10$ episodes, in episodes) are for r2+r3. Full: RRSI's own decision. Best value per column in bold.

adjusted for multiplicity. IL's registration reported 90% bootstrap intervals for its three primary outcomes; with 95% Welch intervals and Holm's adjustment over the three, the transfer difference is -4.4 $[-9.6, +0.7]$ (Holm $p = 0.11$), the cost difference $-\$0.35$ $[-0.70, +0.01]$ ($p = 0.11$) and the episode difference -342 $[-562, -121]$ ($p = 0.02$; `verify/posthoc_r14.py`).

E Reproducibility

The code, environments, intervention banks, registrations and analysis scripts are submitted as anonymised supplementary material, with every candidate's evidence and held-out episodes. The agent is DeepSeek-V4-Flash and the analyst, proposer, critic and judge DeepSeek-V4-Pro, through the provider's Anthropic-compatible endpoint; Study 2 runs at temperature 0. Study 2's tables are produced by the modules of `verify/` (`analyze`, `e1r`, `posthoc_*`, `cl`, `il`, `bk`); Appendix S24 maps every table and figure to its script.

			Full evaluation's acceptances			D(rule) − D(full), points per round			
Rule	Recall [90%]	Acc.	Gain kept	Gain dropped	Dropped	Forgone	Avoided	Other	Total
Sequential full	0.74 [0.67, 0.81]	0.00	+0.11	+0.30	5.4	−0.06	+0.04	−0.00	−0.02
Sequential sample@80	0.48 [0.39, 0.56]	0.11	+1.06	−0.68	10.9	−0.10	+0.19	−0.07	+0.03
Sample@80	0.76 [0.66, 0.85]	0.21	+0.81	−1.94	5.0	−0.03	+0.15	−0.12	+0.00
Sample@40	0.55 [0.48, 0.62]	0.28	+0.60	−0.38	9.4	−0.11	+0.16	−0.14	−0.10
Net@40	0.54 [0.48, 0.61]	0.34	+0.27	+0.03	9.6	−0.12	+0.12	−0.13	−0.14
Replay−null@40	0.23 [0.10, 0.35]	0.15	−0.32	+0.30	16.3	−0.28	+0.21	−0.37	−0.43
LLM judge	0.48 [0.30, 0.65]	0.76	−0.12	+0.42	11.0	−0.21	+0.16	−0.38	−0.44
None	0.62 [0.55, 0.69]	0.88	+0.24	+0.03	8.0	−0.11	+0.11	−0.51	−0.51
Keep incumbent	0.00 [0.00, 0.00]	0.00	–	+0.16	21.0	−0.32	+0.28	+0.00	−0.04
Sequential full, r1 (5)	0.69 [0.49, 0.90]	0.00	+1.52	+2.27	1.5	−0.09	+0.00	+0.02	−0.07
Sequential full, r1–r3 (26)	0.73 [0.67, 0.80]	0.00	+0.37	+0.73	7.0	−0.07	+0.03	+0.01	−0.04
Sequential full, r4 (2)	0.74 [0.49, 1.00]	0.00	+5.57	+1.70	0.5	−0.09	−0.00	+0.02	−0.06
Sequential full, E3 (3)	0.88 [0.64, 1.00]	0.00	+0.59	−2.60	0.4	−0.00	+0.05	−0.00	+0.05

Table 6: Agreement with full evaluation's acceptances on r2+r3 (21 acceptances in 80 rounds; post hoc). Recall: expected share of full evaluation's acceptances the rule also makes, with a 90% block-bootstrap interval. Acc.: share of the rounds in which full evaluation keeps the incumbent where the rule accepts a candidate. Gain kept and dropped: mean held-out gain (points) of full evaluation's accepted candidates, weighted by the probability that the rule makes or does not make the same acceptance; Dropped: expected number it does not make. The last four columns split the difference in decision value: gains forgone by dropping helpful acceptances, harm avoided by dropping harmful ones, the value of what the rule accepts instead (another candidate, or one in a round where full evaluation keeps the incumbent), and the total. Lower block: sequential full evaluation on the other trajectories, with full evaluation's number of acceptances in parentheses.

	r2+r3		r2+r3, final ΔC		Airline r1–r4		AppWorld		Live loops	
Predicted p	mean p	admitted	mean p	admitted	mean p	admitted	mean p	admitted	mean p	admitted
$[0.00, 0.01)$	0.003	0.08	0.003	0.00	0.003	0.10	0.004	0.03	0.004	0.22
$[0.01, 0.05)$	0.028	0.08	0.028	0.02	0.026	0.14	0.030	0.03	0.027	0.08
$[0.05, 0.20)$	0.12	0.12	0.12	0.07	0.11	0.17	0.10	0.05	0.12	0.18
$[0.20, 0.50)$	0.34	0.21	0.34	0.19	0.33	0.25	0.32	0.10	0.35	0.16
$[0.50, 0.80)$	0.64	0.36	0.64	0.39	0.65	0.38	0.65	0.38	0.65	0.51
$[0.80, 0.95)$	0.88	0.52	0.88	0.62	0.88	0.52	0.89	0.66	0.87	0.64
$[0.95, 1]$	0.98	0.78	0.98	0.89	0.99	0.74	0.98	0.84	0.98	0.74
Stopped: mean p / admitted	0.021	0.08	–		0.014	0.12	0.025	0.03	0.018	0.13
Stopped: selected by full	0.07		–		0.10		0.02		0.10	
Admitted by full: stopped	24% of 24		–		28% of 23		15% of 5		35% of 23	

Table 7: Calibration of sequential full evaluation's predicted admission probabilities (post hoc). Rows bin the interim looks by p; "admitted" is the share of those looks whose candidate full evaluation admitted. Recorded trajectories: 400 orders per candidate. Live loops: cl1, cl2 and s1–s4, outcomes from the loop or its shadow evaluation. "Final ΔC": p recomputed with the complete evaluation's cost change (a diagnostic). Bottom rows: stopped candidates (mean p at the stop, share admitted and selected by full evaluation) and the share of full evaluation's admitted candidates that were stopped.

Conclusion	Evidence	Specified	Status with adjusted inference
Few measured candidates measurably helped held out	of 183, unadjusted 95% intervals above zero for 8 (4–5 expected by chance) and below for 12, 163 unresolved; mean gains on r1–r3 -1.5, -1.25, $+0.14$ points	registered deployments; split post hoc	descriptive, no test
Evidence that never runs solved tasks accepted harmful changes	the judge and replay−null accepted the write-blocking candidate; the judge and replay without null replays the crashing one	registered collection; two cases	case evidence, no test
No rule was shown to outperform keeping the incumbent	r2+r3, every rule	registered	holds (no interval excludes zero)
The reference gains mainly in early airline rounds	D(full) $+0.68$ $[+0.08, +1.57]$ on airline rounds 0–9 of r1–r3, -0.13 to -0.07 in its other cells	post hoc	descriptive split; other cells' intervals are wide
Cheaper rules are non-inferior on recorded decisions	sample@40, net@40 (Holm $p = 0.045$; 0.056 with task draws shared across trajectories); sequential rules ($p < 0.001$)	registered, Holm	holds for the sequential rules; borderline for the @40 rules (0.045–0.056 across dependence structures and bootstrap sizes); keeping the incumbent passes too (post hoc)
The sequential rule drops full evaluation's acceptances	recall 74% on r2+r3 (post hoc); IL's loops made 14 of 20; BK's rule stopped full evaluation's choice in 7 of 16 rounds	registered shadow evaluations (descriptive)	descriptive, no test
Its stops are overconfident on RRSI's cost branch	stopped at mean probability 0.02, admitted in 8% and 13% (r2+r3, live; 95% intervals exclude the prediction), 3% on AppWorld	post hoc	on τ^2 and in live loops, not detectably on AppWorld; absent without that branch; its fix, the cost-aware rule, made full evaluation's choice in all 40 BK rounds (registered, descriptive)
Loops it drives transfer less	IL -4.4 (95% $[-9.6, +0.7]$); BK -0.8 (95% $[-3.4, +1.7]$, Bonferroni $[-4.1, +2.4]$)	registered	IL: non-inferiority at 5 points not established, loss not shown at 95%; BK: difference unresolved, a five-point loss excluded (post hoc reading; no registered non-inferiority test)
The dropped admissions cost transfer	BK admission effect -1.6 $[-3.0, -0.2]$ at 90%	registered reading (unadjusted)	not established: Holm $p = 0.19$, 95% interval includes zero; -1.0 to -2.2 without one block; 61% power at four blocks
The proposer's history matters	BK history effect $+0.7$ on 1.4 df	registered	unresolved
A calibration imbalance contributed to IL's loss	band-adjusted IL -2.1; offline band sweep	post hoc	not isolated
A cost-aware rule keeps the acceptances	88% recall offline (designed on these rounds); BK: full evaluation's choice in all 40 rounds	registered for BK (descriptive)	descriptive; its gains unresolved
Evaluation saves a small share of loop cost	episodes -39% and -37%, cost -10% and -12% (IL, BK)	IL primary; BK descriptive	episode saving holds ($p = 0.02$); IL's cost saving not shown at 95%

Table 8: Conclusions of Study 2 by the evidence behind them. Specified: whether the analysis was registered before its episodes ran. With adjusted inference: what remains with registered analyses only, 95% intervals and Holm's adjustment within each registered family.